

KHYAGT, Russian FederationWMO: **309250**Lat: **50.364N**Lon: **106.452E**Elev: **2616**StdP: **13.36**Time Zone: **8.00 (E08)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.4	-19.9	-28.1	1.3	-22.8	-24.7	1.5	-19.3	15.8	4.9	12.2	6.9	0.4	20	0.582

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.0	86.9	65.9	83.3	64.2	79.9	62.9	68.5	82.6	66.2	79.4	64.4	76.5	6.0	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
63.6	96.9	75.2	61.5	90.0	72.5	59.7	84.2	70.9	34.5	82.7	32.5	79.7	31.1	76.5	77.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.5	14.3	12.1	-26.8	93.3	5.4	4.4	-30.7	96.5	-33.9	99.1	-37.0	101.5	-40.9	104.7		
(5)			-26.8	71.4	5.4	2.7	-30.7	73.3	-33.9	74.9	-36.9	76.4	-40.8	78.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	34.2	-3.6	4.4	21.8	39.0	50.9	63.2	67.9	63.1	50.6	34.7	15.8	0.6
	DBStd	26.30	9.00	10.36	11.27	9.12	9.15	7.74	6.26	6.60	8.07	9.03	11.17	9.63
	HDD50	7390	1661	1276	876	346	102	4	0	2	90	478	1025	1530
	HDD65	11511	2126	1696	1341	781	444	125	39	115	434	940	1475	1995
	CDD50	1617	0	0	0	15	131	399	554	407	108	3	0	0
	CDD65	265	0	0	0	0	8	70	128	56	3	0	0	0
	CDH74	2464	0	0	0	3	141	716	1130	451	23	0	0	0
	CDH80	702	0	0	0	0	29	204	358	109	2	0	0	0
(14) Wind	WSAvg	4.2	2.3	3.0	4.5	5.7	6.3	5.4	4.4	4.3	4.5	4.0	3.3	2.3
(15) Precipitation	PrecAvg	13.8	0.2	0.2	0.2	0.6	0.8	2.6	3.4	3.1	1.7	0.6	0.3	0.2
	PrecMax	24.3	0.6	0.4	0.8	2.3	1.9	7.5	8.3	9.1	3.1	1.6	0.7	0.5
	PrecMin	8.5	0.0	0.0	0.0	0.0	0.2	0.3	0.7	1.1	0.4	0.0	0.0	0.0
	PrecStd	3.4	0.1	0.1	0.2	0.5	0.5	1.6	1.8	1.7	0.8	0.4	0.2	0.1
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.0	32.9	54.9	70.5	84.7	90.5	93.0	88.3	77.1	62.6	44.7	26.1
		MCWB	19.7	26.8	41.3	51.0	60.1	66.4	70.9	68.3	60.9	48.7	36.1	22.4
	2%	DB	17.2	27.2	47.0	64.4	78.2	85.2	88.0	82.8	72.0	56.7	39.4	20.7
		MCWB	14.6	23.1	37.1	47.7	57.1	63.4	67.3	65.4	58.0	44.5	32.7	17.8
	5%	DB	12.5	23.0	42.2	59.4	73.4	82.0	84.2	79.0	68.1	52.0	35.4	17.1
		MCWB	10.4	19.8	33.8	44.8	54.7	62.2	65.9	63.5	55.2	42.1	30.4	14.7
	10%	DB	8.8	19.3	37.9	54.8	68.5	77.9	80.4	75.2	63.9	47.6	31.5	13.9
		MCWB	7.1	16.7	30.9	42.2	52.0	60.8	64.4	61.8	53.1	39.3	27.4	12.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.7	27.1	41.9	51.9	61.9	68.8	73.1	70.0	63.0	49.0	36.7	22.3
		MCDB	22.5	32.2	54.3	68.3	82.0	85.6	89.8	85.0	74.8	61.4	43.9	25.7
	2%	WB	14.6	23.0	37.5	48.3	58.3	65.9	69.8	66.7	59.2	45.3	33.2	18.0
		MCDB	17.0	27.0	46.7	62.9	75.9	81.4	84.3	80.0	69.1	55.9	38.9	20.7
	5%	WB	10.5	19.7	34.2	45.3	55.5	63.8	67.4	64.6	56.5	42.4	30.3	14.9
		MCDB	12.4	22.8	41.7	58.5	72.0	78.1	80.1	76.2	66.1	51.0	35.3	17.1
	10%	WB	7.1	16.7	31.1	42.5	52.7	62.0	65.5	62.7	53.9	39.7	27.6	12.0
		MCDB	8.7	19.2	37.6	54.4	67.2	75.3	77.5	73.0	63.0	47.0	31.4	13.9
(35) Mean Daily Temperature Range	MDBR		13.3	17.0	19.3	20.3	22.0	20.2	18.0	17.6	18.9	17.6	15.1	13.1
	5% DB	MCDBR	14.8	18.9	22.1	27.4	29.1	26.3	23.8	23.3	23.7	23.4	17.7	14.5
		MCWBR	12.9	15.4	14.5	16.3	16.0	12.2	10.2	10.7	12.2	14.0	12.2	12.5
	5% WB	MCDBR	14.6	18.6	21.4	26.1	27.6	22.6	20.2	20.8	21.5	22.4	17.2	14.6
		MCWBR	12.7	15.1	14.2	16.0	15.7	11.5	9.8	10.5	11.3	13.8	12.2	12.5
(40) Clear-Sky Solar Irradiance	taub		0.257	0.279	0.317	0.363	0.412	0.414	0.403	0.372	0.323	0.297	0.276	0.259
	taud		2.252	2.237	2.262	2.250	2.130	2.237	2.313	2.382	2.468	2.467	2.340	2.227
	Ebn at Noon		249	272	281	279	268	268	269	273	278	266	239	226
	Edh at Noon		22	29	35	39	46	42	38	34	28	23	20	19
(44) All-Sky Solar Radiation	RadAvg		502	876	1322	1611	1918	1984	1838	1570	1255	819	506	375
	RadStd		17	25	63	83	107	94	107	98	100	50	23	24

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65